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Dependence of Mineral Dissolution Rates on Two-Phase Hydrodynamics in Single Pores

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Key Points:

- Benchmark experiments and pore-scale modeling of calcite dissolution in single pores in two-phase flow
- Internal circulation enhances transport in the aqueous phase and thus surface area averaged dissolution rate
- Effective surface area should not be simply corrected using the average saturation

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Multiphase flow and mineral reactions co-exist in many subsurface systems, and thus elucidating and quantifying the impacts of multiphase flow on mineral reaction rates are crucial. Our study established a benchmark experiment on calcite dissolution in cylindrical pores in slug flow, and performed a series of simulations using a validated pore-scale multiphase reactive transport model. The results revealed contrasting impacts of the two-phase flow on transport in the aqueous phase and the effective surface area, and that the relative contribution of these factors determines whether the bulk reaction rate in two-phase flow is lower (regime I) or higher (regime II) than that in single-phase flow. Scaling laws were attempted to relate reaction rate with velocity to account for the enhanced transport by internal circulations, and variability in the effective surface area was also discussed. This study provides useful insights for upscaling mineral reaction rate in two-phase flow in porous media.

Plain Language Summary Mineral dissolution in multiphase flow is widely observed in environmental and energy systems, but whether or how it differs from the single-phase flow environment is unclear. Large scale predictions typically either assume that multiphase flow does not affect mineral dissolution or a power law function of saturation can be used to correct for the effect on surface area and thus reaction rate. In our study, we used a tractable single pore system to investigate the impact of two-phase flow with alternating gas bubble and liquid droplet on calcite dissolution experimentally and numerically. We highlighted that two-phase flow tends to reduce reaction rate by lowering surface area and fluid availability, and increase reaction rate by enhancing solute transport and mixing. The net effect depends on the relative magnitude of the two factors, and is generally an increase in reaction rate at high flow rates in two-phase flow. Relationships between multiphase reaction rate and parameters like velocity and droplet length can be drawn for certain conditions in our simple system, yet more efforts will be needed for porous media in which the two-phase flow behaviors including droplet size are more complex.

1. Introduction

Multiphase flow is widely observed in natural and engineered systems in which geochemical reactions including mineral dissolution are involved (Agaoglu et al., 2015; Ahmadi et al., 2025; Deng et al., 2023; Deng & Spycher, 2019; Jiménez-Martínez et al., 2020a, 2020b; Viswanathan et al., 2022; Xing et al., 2025). Therefore, it is critical to understand the interactions between multiphase flow dynamics and geochemical reactions.

Multiphase flow affects geochemical reactions by dictating the spatial distribution of the aqueous phase and solute transport therein (Jiménez-Martínez et al., 2015, 2020a, 2020b; Rolle & Le Borgne, 2019). It can increase velocities perpendicular to the main flow direction via for example, internal circulations and create flow structures that enhance hydrodynamic dispersion (Ben-Noah et al., 2023; Jiménez-Martínez et al., 2015, 2020a, 2020b; Ma et al., 2015; Mathiesen et al., 2023; Qian et al., 2019; Tice et al., 2003; Velásquez-Parra et al., 2022). It also affects interphase mass transfer when different fluid phases are miscible (Chomsurin & Werth, 2003; Hussain & Gupta, 2025; Niessner & Hassanizadeh, 2009), by controlling interfacial area and internal circulations (Hussain & Gupta, 2025; Soh et al., 2017).

For heterogeneous reactions at the fluid-solid interfaces, multiphase flow controls reaction rates by influencing solute transport and the reactive surface area (Li et al., 2022). Surface area in contact with the aqueous phase and reaction rate are reduced by the non-wetting phase (Harrison et al., 2017; Hwang et al., 2025; Song et al., 2018; Zhou et al., 2023, 2025). In contrast, multiphase flow can accelerate reactions that consume components within

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the gas phase by increasing their availability (Ahmadi et al., 2025; Harrison et al., 2016). Mineral reactions in multiphase flow can reshape the structural evolution of porous media, homogenizing mineral dissolution and thus suppress wormholing that would otherwise form in single-phase flow (Babaei & Sedighi, 2018; Jiménez-Martínez et al., 2020a, 2020b; Ott & Oedai, 2015; Soulaine et al., 2018; Zhang et al., 2024). Geochemical reactions also influence multiphase flow dynamics by consuming or generating gases, and by altering interfacial tension and wettability (Maes & Geiger, 2018).

These complex interplays between multiphase flow and geochemical reactions can be readily observed in experiments and modeled by pore-scale simulators, but they are rarely accounted for in continuum scale modeling. Efforts that attempt to capture the effects of multiphase hydrodynamics on flow, solute transport, and heterogeneous reaction rate are featured by relating relative permeability (Deng et al., 2023), tortuosity (Millington & Quirk, 1961), and reactive surface area (Sonnenthal et al., 2005) with water saturation using empirical power law relationships. However, they lack pore-scale mechanisms and require system-specific parameterization.

Developing mechanistic constitutive relationships requires fundamental understanding of multiphase reactive transport processes. Investigations of transport behaviors in two-phase flow typically focus on conservative solutes or aqueous reactions (Ben-Noah et al., 2023; Jiménez-Martínez et al., 2015, 2020a, 2020b), whereas surface reactions may cause deviations (Hayes & Kolaczkowski, 1999). Studies that involve heterogeneous reactions, however, tend to emphasize the evolution of fluid-solid interfaces (Hwang et al., 2025; Jiménez-Martínez et al., 2020a, 2020b), and thus provide limited constraints on the effect of transport and surface area. Furthermore, diverging observations have been reported, showing both higher and lower mineral reaction rate in two-phase flow (Jiménez-Martínez et al., 2020a, 2020b; Li et al., 2022; Zhang et al., 2024). Therefore, tractable experiments and simulations are still in need to identify and quantify the underlying mechanisms.

Motivated by this research need, our study aims to unravel the impacts of two-phase flow on mineral dissolution rate in a well-defined system. Through a benchmark experiment of two-phase slug flow—aka Taylor flow—with stable alternating bubbles and droplets in Iceland spar channels, we observed both regimes when the reaction rate in two-phase flow is higher and lower than that in the single-phase flow. Furthermore, pore-scale modeling benchmarked by the experimental data enabled systematic analyses of two-phase flow dynamics, mass transfer in the liquid slug, and calcite reaction rate. Furthermore, we discuss the implications for upscaling mineral reaction rate in two-phase flow systems.

2. Methodology

2.1. Experiments and Materials

Experiments were performed to quantify calcite dissolution rate in a single circular channel under single- and two-phase flow conditions (Figure 1a). Cylindrical channels were drilled through polished single crystals of calcite (Iceland spar) to represent single pores (Li et al., 2008). Fluidic ports were glued on the crystal surfaces using waterproof epoxy resin adhesive (Devcon No. 14250) and connected to FEP tubing. Three reactive channels were prepared, with length (L) and diameter (d) being 9.1–1.4 mm, 12.09–1.8 mm, 8.93–1.3 mm for channel-1, -2, and -3, respectively. The measured contact angle on calcite surface is 60° (Figure S1 in Supporting Information S1). Two peristaltic pumps (LC-TFP-YZ15A) were used to inject the solution and air into the channel. A three-port T junction manifold was used to generate a stable slug flow. The outlet of the system is open to the atmosphere. The aqueous phase is a 10 mmol/L NaCl solution with pH adjusted to 5 using HCl, which prevents CO_2 degassing and geometrical alteration during the experiments. Experiments were performed at flow rates (q) between 10 μL/min and 10 mL/min for both single-phase (water) and two-phase (water and air at an equal ratio) flow in each channel (Table S1 in Supporting Information S1). The Reynolds number ($Re = \rho_w U/\mu_w$) and Capillary number ($Ca = \frac{\mu_w U}{\sigma}$)—calculated from water density (ρ_w), water viscosity (μ_w), total average velocity (U), and surface tension (σ)—ensure the laminar flow regime and relevance to typical geologic systems (Mathiesen et al., 2023). For each experiment, flow through the reactive channel was recorded by a Sony camera (Sony α7 IV, 90 mm lens), and three 5 mL liquid samples were collected for calcium concentration ($C_{Ca,out}, [\mathrm{mol}/L]$) measurements using ICP-OES (ThermoFisher iCAP PRO) and ICP-MS (ThermoFisher iCAP TQ) for experiments at flow rates ≥5 mL/min. The bulk reaction rate ($R, [mol/s]$) is calculated using Equation 1, in which $C_{Ca,in}$ is the influent calcium concentration.

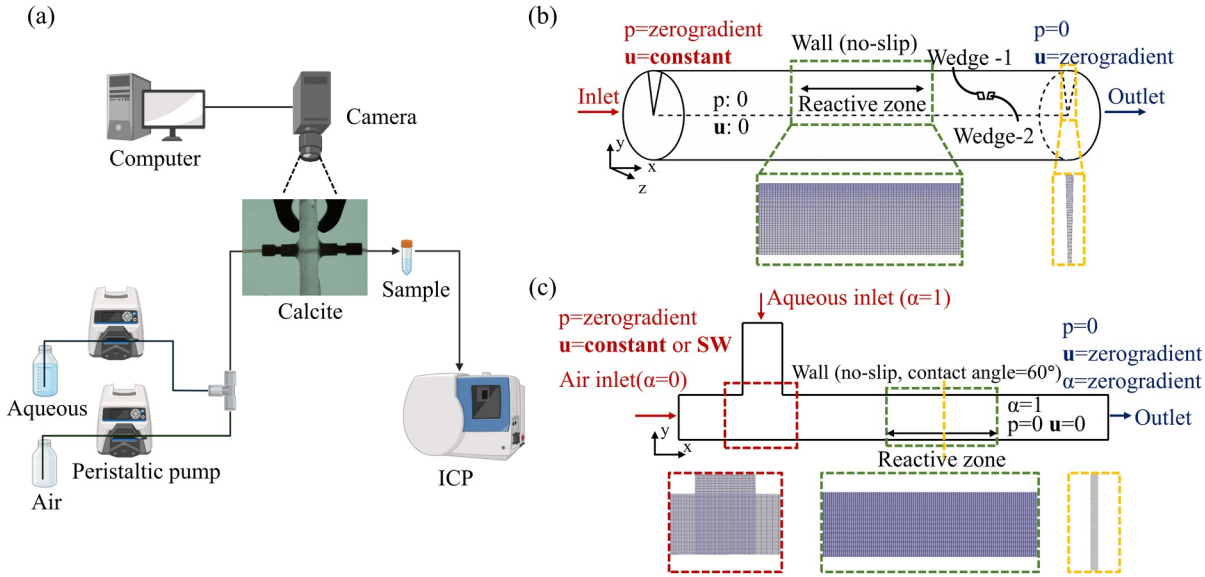


Figure 1. (a) Schematic of the experimental setup, and illustration of the geometry and meshing used in the (b) single-phase and (c) two-phase flow reactive transport simulations.

$$R = q(C_{Ca,out} - C_{Ca,in}) \tag{1}$$

2.2. Numerical Modeling

2.2.1. Model

The pore-scale simulations were performed using CrunchFOAM (Li et al., 2022), which couples OpenFOAM (Jasak, 2009) and CrunchTope (Steefel et al., 2015) via the alquimia interface (Molins et al., 2025).

Two-phase flow is solved using the extensively validated OpenFOAM solver interFoam (Deshpande et al., 2012; Ferrari et al., 2015; Pavuluri et al., 2018; Shuard et al., 2016), which implements the modified volume of fluid (VOF) method (Hirt & Nichols, 1981), as described by Equations 2–4.

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \left[\nabla \cdot \left(\mu\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)\right)\right] + \rho g + \mathbf{F}_{st} \tag{2a}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{2b}$$

where $\mathbf{u}$ is the velocity vector, and $\mathbf{F}_{st}$ is the surface tension force. Fluid density ρ and viscosity μ are calculated using Equation 3 as weighted average of the wetting (w) and nonwetting (nw) phases, and α is the volume fraction of the wetting phase, which is updated via Equation 4 with $\mathbf{u_r}$ being the relative velocity between the two phases.

$$\rho = \rho_w \alpha + \rho_{nw}(1-\alpha) \tag{3a}$$

$$\mu = \mu_w \alpha + \mu_{nw}(1-\alpha) \tag{3b}$$

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) + \nabla \cdot (\alpha(1-\alpha)\,\mathbf{u_r}) = 0 \tag{4}$$

Transport of the primary species (j) is solved using the compressive Continuous Species Transfer (c-CST) method (Maes & Soulaine, 2018; Equation 5).

$$\frac{\partial C_j}{\partial t} + \nabla \cdot (\mathbf{u} C_j) = -\nabla \cdot \left(\frac{(1-H_j)\,C_j}{\alpha + (1-\alpha)\,H_j}\alpha(1-\alpha)\,\mathbf{u_r}\right) + \nabla \cdot \left(\widehat{D_j}\nabla C_j + \Psi_j\right) \tag{5a}$$

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$$\widehat{D_j} = \alpha D_{j,w} + (1-\alpha) D_{j,nw} \tag{5b}$$

$$\Psi_j = -\widehat{D_j} \frac{1 - H_j}{\alpha + (1-\alpha) H_j} C_j \nabla \alpha \tag{5c}$$

where $D_{j,w}$ and $D_{j,nw}$ are the diffusion coefficient in the aqueous and non-wetting phase, and set to 1×10^{-9} m^2/s and 1×10^{-5} m^2/s, respectively. Ψ_j represents the flux across the fluid-fluid interface, and H_j is Henry's coefficient, which is set to 1×10^{-12} as interphase mass transfer is negligible in our study.

Following the operator splitting approach, concentration fields of the primary species calculated from Equation 5 are used to solve the concentrations of all secondary species based on the law of mass action, which is equivalent of treating homogeneous reactions in the aqueous phase as equilibrium reactions. Mineral reactions, are then solved kinetically using Equation 6:

$$\frac{\partial C_j}{\partial t} = R_j \tag{6a}$$

$$R_{Ca} = R_C = A_{geo} r \tag{6b}$$

$$r = (k_1 a_{H^+} + k_2 a_{H_2CO_3} + k_3)\left(1 - \frac{a_{Ca^{2+}} a_{CO_3^{2-}}}{K_{eq}}\right) \tag{6c}$$

where k_1, k_2, and k_3 are reaction rate coefficients [$mol/m^2/s$] for the three parallel reactions of calcite (Chou et al., 1989), and A_{geo} is the geometric surface area $[m^2]$ in pore-scale models, K_{eq} is the equilibrium constant, and a_{H^+} and $a_{H_2CO_3}$ are the activities of proton and carbonic acid, respectively. By using constant rate coefficients and smooth geometries in the simulations, we assumed no reactivity variability or surface roughness or coupling between them. All thermodynamic and kinetic parameters are documented in Tables S3 and S4 in Supporting Information S1.

2.2.2. Simulation Setup

In the simulations, the reactive channel is patched with a 10 mm non-reactive zone respectively at the upstream and downstream to allow flow development and stabilization. In single-phase flow simulations, given the axisymmetric nature of the cylindrical channel, a wedge of 5° was modeled (Figure 1b). In two-phase flow, a 2D slice of a parallel-plate channel was simulated (Figure 1c), because the T-junction used to create the water-gas slug flow cannot be readily represented by the wedge setup. For the same reason, no boundary layers were added in the two-phase flow meshes. The channel width is discretized into 50 grid cells ($d = 50\Delta y$). Mesh convergence tests and the comparison between a full 3D simulation and the corresponding 2D simulation are detailed in SI. The differences are within 5%, which justifies our modeling decision to balance accuracy and computational cost.

Fluid properties and boundary conditions used in the simulations are summarized in Tables S5 and S6 in Supporting Information S1. As the internal structure of the experimental T-junction manifold can not be resolved, two types of inlet boundary conditions were devised for the two-phase simulations, to capture different features of the experimental setup. In one case (referred to as TJ), the water inlet ($\alpha = 1$) and gas inlet ($\alpha = 0$) have the same uniform velocity ($u_{w,inlet} = u_{nw,inlet} = U$), and the bubble/droplet size is controlled collectively by the flow velocity, fluid properties and the given T-junction geometry as in the experiments. In the second case (referred to as SW), the velocity at the inlets are set as square wave like periodic functions (Equations 7a and 7b), and the bubble/droplet size is specified independently via the period (t_c) and can be set exactly the same as the experiments.

$$u_{w,inlet}(t) = \begin{cases} U, \sin(2\pi t/t_c) > 0 \\ 0, \sin(2\pi t/t_c) < 0 \end{cases} \tag{7a}$$

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$$u_{nw,inlet}(t) = \begin{cases} 0, \sin(2\pi t/t_c) > 0 \\ U, \sin(2\pi t/t_c) < 0 \end{cases} \quad (7b)$$

The TJ case includes the upstream control on two-phase flow dynamics, whereas the SW case can represent scenarios when the upstream geometry and flow dynamics are complex and unknown. Three additional channels (4–6) with varied lengths (100 μm–10 mm) and diameters (100 μm–1 mm) were simulated in the SW case. All simulation conditions are summarized in Table S2 in Supporting Information S1. The simulations ran until 10 pore volumes were injected, and the steady state is observed after five pore volumes in all cases.

To examine the reaction rate from the simulations, the instantaneous bulk reaction rate ($R_p(t)$) and surface area averaged reaction rate ($r_p(t)$) within the reactive domain were calculated using Equation 8.

$$R_p(t) = \sum_{n=1} r_n(t)\, V_n(t)\alpha_n(t) \quad (8a)$$

$$r_p(t) = \frac{\sum_{n=1} r_n(t)\, V_n(t)\alpha_n(t)}{\sum_{n=1} A_n(t)\beta_n(t)} \quad (8b)$$

where $p = t, s$ for two-phase and single-phase flow simulations, respectively. $r_n(t)$ [mol/m^3/s], $V_n(t)$ [m^3], and $A_n(t)$ [m^2] are the reaction rate, volume and wall surface area in grid cell n, and α_n^t and β_n^t are local water saturation in the grid cell and on the wall at time t. To compare with the experiments, the steady state reaction rates were calculated from data during the second half of the simulations using Equation 9, and the exact time points (t_1 and t_2) defining this duration varies slightly among simulations to ensure that complete water/gas slug cycles are included.

$$R_p = \int_{t_1}^{t_2} R_p(t)\, dt/(t_2 - t_1) \quad (9a)$$

$$r_p = \int_{t_1}^{t_2} r_p(t)\, dt/(t_2 - t_1) \quad (9b)$$

3. Results and Discussion

3.1. Single-Phase Versus Two-Phase Reaction Rate

For all channels, the single-phase bulk reaction rate (R_s) increases with the flow rate (Figure 2a). At the same flow rate, the reaction rate is higher for channel-2, which has larger length, diameter, and reactive surface area. However, the surface area averaged reaction rate (r_s) is lowest for channel-2, and is highest in general for channel-3 that has the smallest diameter (Figure S1b in Supporting Information S1). This is because the smaller diameter would reduce the transport time across the channel.

The simulations successfully reproduced the experimental measurements, with relative differences <10% across all simulations (Figure 2c). The simulated velocity and concentration fields show a clear parabolic velocity profile and well-developed concentration gradients (Figure 3a). The hydrodynamic and concentration boundary layer becomes thinner as velocity increases (Figures 3a and 3c), suggesting a decreasing transverse transport limitation. These observations explain why R_s and r_s increases with flow velocity and suggest that at the flow rates studied here the reactions are transport controlled.

The reaction rate range and overall dependence on flow rate and channel diameter are consistent with the literature (Elkhoury et al., 2013; Li et al., 2008). While reactivity variability and roughness at the finer scale may be present in the experiments, the good agreement between simulations and experiments confirm that they do not affect the averaged reaction rates notably under the conditions (i.e., transport-controlled regime) studied here in contrast to well-stirred surface-controlled systems (Noiriel et al., 2020). This is consistent with the previous pore-scale

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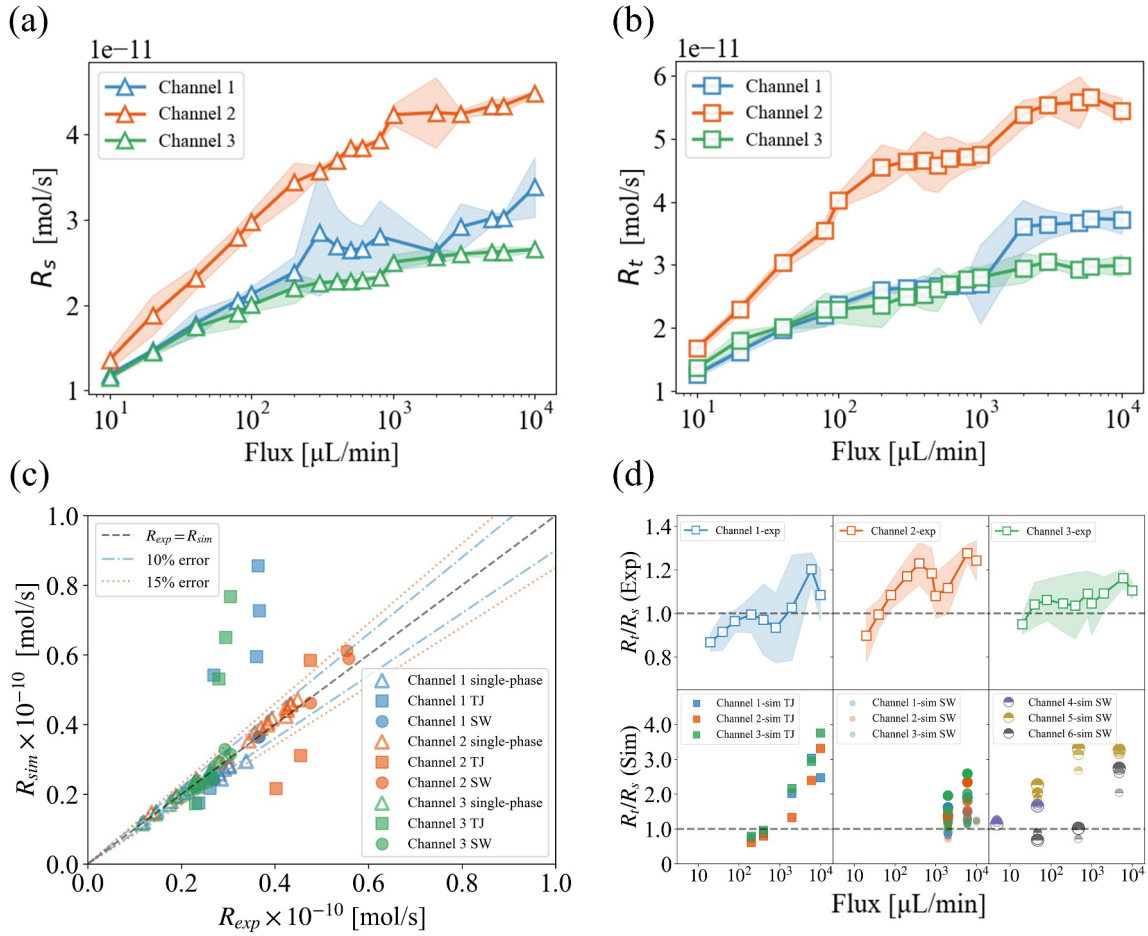


Figure 2. Experimental results of bulk reaction rate in (a) single-phase and (b) two-phase flow, (c) comparison of experimental and simulated bulk reaction rates, and (d) ratio of bulk reaction rates between two-phase and single-phase flow at the same total flow rate. Markers with reduced size and color saturation in (d) denote simulations with smaller slug length (see Table S2 in Supporting Information S1). The shaded areas in (a) and (b) show one standard deviation (σ_{Rt} and σ_{Rs}) of the effluent measurements, and in (d) it shows the standard deviation of R_t/R_t defined as $\sigma_{\text{ratio}} = \frac{R_t}{R_s} \times \sqrt{\left(\frac{\sigma_{Rt}}{R_t}\right)^2 + \left(\frac{\sigma_{Rs}}{R_s}\right)^2}$.

simulations showing minimum impact from low-level roughness (Deng et al., 2018) and with the study using the same sample preparation procedure (Li et al., 2008).

In two-phase flow, the bulk reaction rate (R_t) also increases with water (total) flow rate (Figure 2b), and the variations across channel geometries show a similar trend as those in single-phase flow. The reaction rates predicted by the T-junction simulations using the experimental channel geometries are notably different from the experimental data, because the simulated droplet sizes (2.8–7.5 mm) are significantly lower than those observed experimentally (7–36 mm, Figure S4 in Supporting Information S1). These discrepancies are expected, as the slug size (L_s) is sensitive to the geometry of the T-junction (K. Wang & Luo, 2017) and to the relative orientation of the water and gas phase inlets (Aoki et al., 2011), neither of which can be accurately recreated in the modeling. However, SW simulations that specify L_s as comparable to the corresponding experimental measurements, can reproduce the experimental R_t with differences <15% (Figure 2c).

The experimental measurements showed two regimes regarding the relative magnitude of reaction rate in two-phase and single-phase flow (Figure 2d). In regime I, that is, at flow rates of ~10 μL/min R_t is lower than R_s for all channels regardless of the measurement uncertainties. In regime II, that is, at flow rate larger than ~100 (1,000)μL/min for channel-2 (−1 and −3), R_t is notably larger than R_s. For channel-1 and -3 a wide transition region exists with R_t/R_s varies around 1 considering the measurement uncertainties. While the absolute values of

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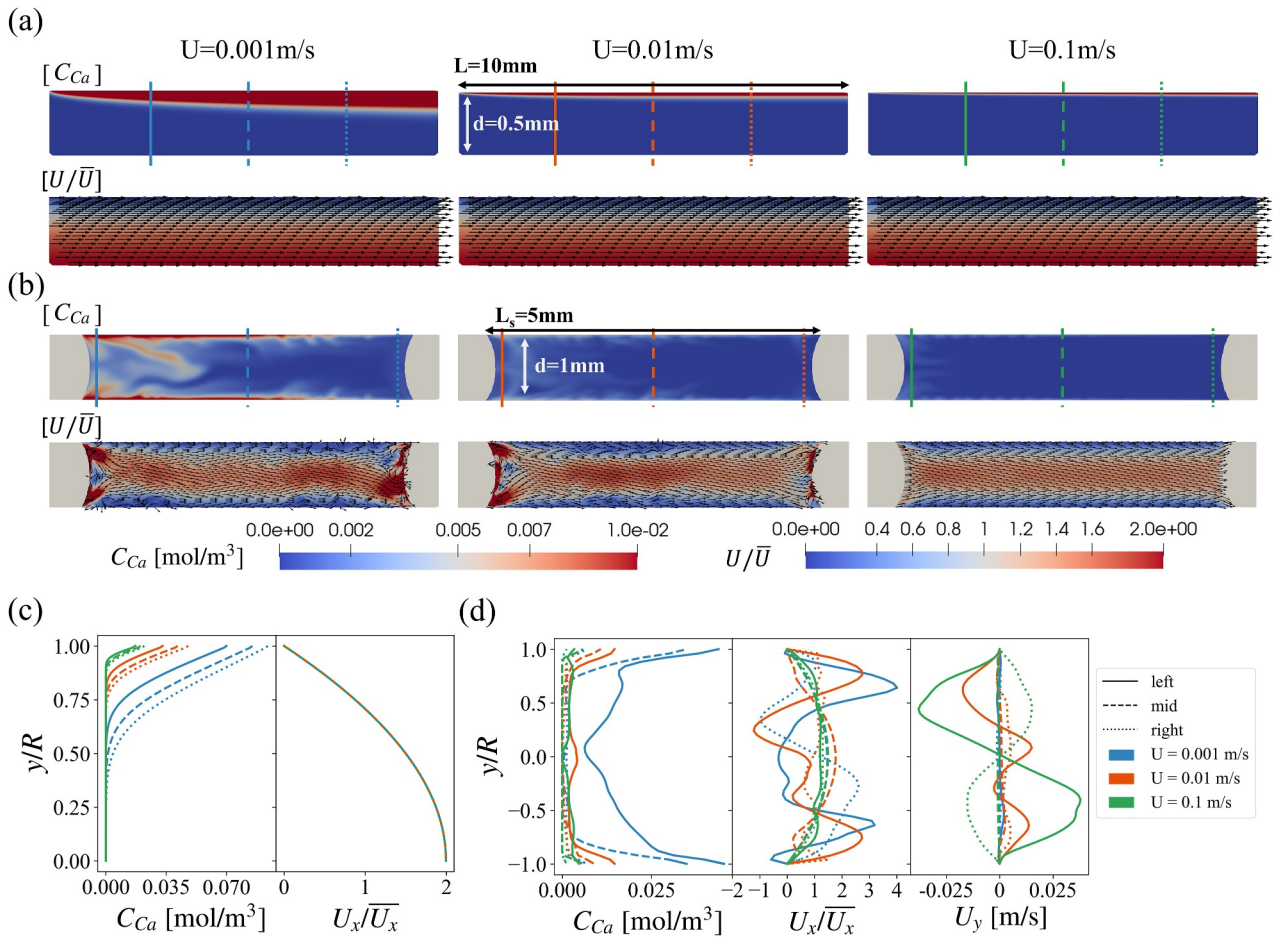


Figure 3. Colormaps of calcium concentrations and velocity fields of (a) single-phase and (b) two-phase simulations at the same total flow rate, in the channel with $L = 10$ mm and $d = 1$ mm at $U = 0.001, 0.01, 0.1$ m/s, and the slug length $L_s = 5$ mm in the two-phase flow. Vertical profiles of concentrations and normalized velocities of (c) single-phase and (d) two-phase simulations at locations highlighted in (a) and (b).

the T-junction simulations are different from the experiments, the T-junction simulations successfully reproduced the two regimes (Figure 2d).

At the same total flow rate, the amount of reactive fluid and the capacity for calcite dissolution are reduced, and the surface area in contact with the reactive fluid over a given time period is smaller (Li et al., 2022). Both factors would contribute to a lower R_t. The presence of regime II indicates the play of another mechanism, that is, enhanced transport perpendicular to the flow direction and thus transverse solute mixing (Jiménez-Martínez et al., 2015, 2020a, 2020b; Tice et al., 2003; Velásquez-Parra et al., 2024). This is confirmed by the velocity and concentration profiles from the two-phase flow simulations (Figures 3b and 3d). Internal circulations are formed in the aqueous slug, especially close to the water-air interfaces. Consequently, velocity in the transverse direction is non-zero, and diffusion is no longer the only transverse transport mechanism as in the single-phase flow. Accordingly, the reaction products, for example, Ca^{2+}, are transported away from the mineral wall, sustaining the thermodynamic driving force (i.e., lower Ca concentration) at the mineral interface. The concentration profiles show more complex gradients instead of the well-developed boundary layer due to the more variable velocity field.

While the internal circulations appear to be more well-developed on the velocity maps at low flow rates, the intensity as measured by the absolute value of the transverse velocity (U_y) is larger at higher flow rates. This explains the more significant enhancement in R_t with respect to R_s at the higher flow rate, when the two-phase flow accelerates transport enough to compensate for the reductions in solution availability and surface area.

The extent of the enhancement from the internal circulation varies even at the same flow rate. For cases with longitudinal advection time scale smaller than one second, R_t increases as L_s decreases, because the relative

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region affected by the internal circulation is larger (SI). For cases with longer longitudinal residence time, there is no obvious dependence on L_s as the internal circulation has more time to enhance mixing within the liquid slug as it migrates along the reactive channel. This complex dependence of reaction rate on L_s while everything else being equal also explains the non-monotonic increase in the experimental R_t/R_s ratio (Figure 2d)—as the variations in the reaction rate correspond to the variations in L_s—and the larger R_t/R_s ranges in the simulations.

3.2. Implications for Upscaling

To inform upscaling, we explored scaling laws for two parameters: the surface area averaged reaction rate (r), serving as an indicator of the extent of transport limitation, and the ratio between the bulk and surface area averaged reaction rate (R/r), which can be considered as the effective surface area (A_e) for the reaction.

For single-phase flow in circular capillary tubes, theoretical mass transfer coefficient (k_m) or Sherwood number (Sh) is given as a function of Peclet number ($Pe = Ud/D$) in Equation 10 (Montagnaro, 2024), where D is the molecular diffusion coefficient. k_m shows reasonable positive correlations with mass transfer coefficients (k_x, k_y) that are evaluated from the concentration gradient and velocity field in the x and y directions from the simulations using Equation 11. But significant scattering was observed between r_s and k_m since Equation 10 was developed for non-reactive walls.

$$Sh = \frac{k_m d}{D} = 3.66\{1 + 0.095Pe\}^{0.45} \tag{10}$$

$$k_x = \overline{\left[\frac{\boldsymbol{u}C - D\nabla C}{C}\right]_x} \tag{11a}$$

$$k_y = \overline{\left[\frac{\boldsymbol{u}C - D\nabla C}{C}\right]_y} \tag{11b}$$

While the increase of r_s with velocity and the dependence on channel diameter were captured by Equation 10, the effect of the channel length is obscure. Our simulations showed that at comparable flow velocities, r_s clearly decreases as the length and width of the channel increases because the solution in shorter channel is further away from equilibrium as measured by the chemical affinity term in Equation 6c. This observation resonates with previous study concluding that mass transfer behaviors with and without reactions at the wall could be considerably different (Hayes & Kolaczkowski, 1999).

Close examination of the simulation data revealed a statistically significant linear relationship between r_s and $\tilde{\tau}$ in the log-log space (Figure 4a), where $\tilde{\tau}$ is an effective residence time calculated from the longitudinal residence time ($\tau_l = \frac{L}{U}$) and the transverse residence time ($\tau_t = \frac{d^2}{D}$) as $\tilde{\tau} = \sqrt{\tau_l \tau_t}$. Equation 12 details the power law relationship between r_s and $\tilde{\tau}$, with the adjusted R^2 being ~0.86.

$$r_s = 1.29 \times 10^{-6} \tilde{\tau}^{-0.20} = 1.29 \times 10^{-6} \left(\sqrt{\frac{Ld^2}{UD}}\right)^{-0.20} \tag{12}$$

For two-phase flow, among a multitude of semi-empirical relationships (Haase et al., 2016; Kreutzer, 2003), two formulations were selected to calculate the mass transfer coefficients.

$$k_1 = \left(2.4 + \frac{1.5d}{L_s}\right)\left(\frac{dU}{LD}\right)^{0.45}\frac{D}{d} \tag{13a}$$

$$k_2 = \frac{2\sqrt{2}}{\pi}\sqrt{DU/L_s} \tag{13b}$$

The formulation given by Equation 13a is a correlation fitted from CFD simulation results for mass transfer from the liquid phase to the wall for Taylor flow in circular capillaries (van Baten & Krishna, 2005). The formulation of Equation 13b was derived from the penetration model for interfacial mass transfer at the slug caps (van Baten &

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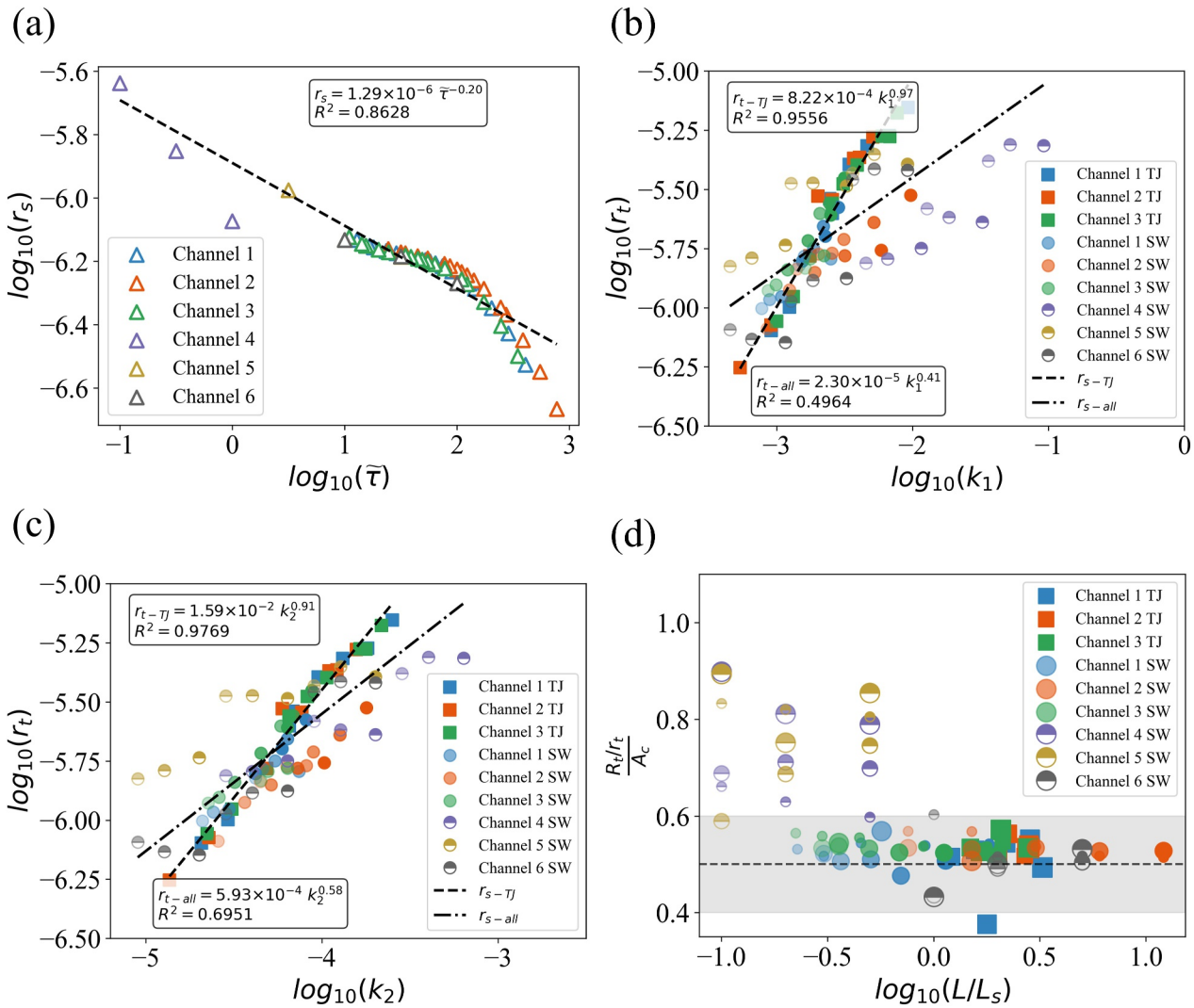


Figure 4. (a) Surface area averaged reaction rate in single-phase flow in relation to $\tilde{\tau}$, surface area averaged reaction rate in two-phase flow in relation to (b) k_1 and (c) k_2, and (d) effective surface area over geometric surface area in relation to L/L_s. Larger markers in (d) represent higher flow rates, and lower color saturation represent smaller slug size.

Krishna, 2004). r_t from the T-junction simulations follow power law relationships with respect to k_1 and k_2 as detailed in Equation 14, with adjusted $R^2 > 0.95$ (Figures 4b and 4c). The interphase mass transfer coefficient can provide a reasonable approximation to mass transfer and reaction rate here, as the underlying mechanism is similar: the interphase mass transfer is also enhanced by the mixing of solute internally and renewal of the solute at the fluid-fluid interphase.

$$r_{t,TJ} = 8.22 \times 10^{-4} k_1^{0.97} \tag{14a}$$

$$r_{t,TJ} = 1.59 \times 10^{-2} k_2^{0.91} \tag{14b}$$

$$r_t = 2.3 \times 10^{-5} k_1^{0.41} \tag{15a}$$

$$r_t = 5.93 \times 10^{-4} k_2^{0.58} \tag{15b}$$

However, when the slug size is independently specified in the SW simulations, the data are more scattered, and the scaling with respect to U reduces from ~1/2 to ~1/4. Adjusted R^2 for the regression between r_t and k_1

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(Equation 15a) and between r_t and k_2 (Equation 15b) for all simulations is ~0.49 and 0.69, respectively. This illustrates that while scaling laws can be developed with high confidence for simple capillary tube systems, they are expected to be more complex in porous media systems. In porous media, the slug size depends on the geometry of the upstream paths and are less predictable, as mimicked by the SW simulations. The scaling laws may still provide some insights as L_s is included, but accurate predictions may be challenging.

This variability that is introduced by simply varying the size of the aqueous slug is also manifested in the effective surface area, defined as $A_e = R_t/r_t$. For cases with $L_s < L$, the ratio between A_e and the actual channel surface area (A_c) is close to the average saturation, which is 0.5. Whereas when L_s is larger, A_e/A_c varies between 0.5 and 0.9.

This highlights an important yet rarely discussed point: temporal upscaling. In the simulations, the actual wetted surface area dictated by water saturation in the pore is used to calculate instantaneous bulk reaction rate (Equation 8). But the time averaged wetted surface area ratio (i.e., saturation) is not necessarily equal to A_e/A_c. Consequently, even when transport is properly upscaled and r_t can be accurately assessed, R_t may be underestimated when the saturation is used to correct for the effect of wetted surface area. This observation also reiterates that correcting reactive surface area as a power law function of water saturation may be problematic, because the exponent may not be constant and would vary with the size of the water clusters and velocities as demonstrated in our study (Figure 4d).

To summarize, conditions that increase transverse velocity and mass transfer and maintain high effective surface area will likely to result in regime II, when $R_t > R_s$. Given that both r_s and r_t scale with U^{η}, where η is empirical and is between 0 and 1 as indicated by our study, it follows that the ratio of r_t/r_s would scale with a power law with respect to U. Future studies that develop such scaling relationships for actual porous media would help devise quantitative criteria that delineate regime I and regime II to more accurately assess the impact of two-phase flow dynamics on mineral reaction rates.

4. Conclusions

We performed benchmark experiments on calcite dissolution in single pores in single-phase and two-phase flow, and pore-scale reactive transport simulations that successfully reproduced the experimental observations. Our results showed that two-phase flow can enhance mixing in the liquid phase via internal circulations and thus elevates r_t. The extent of increase can compensate for the reduction in effective surface area at high flow rates, such that R_t is larger than R_s at the same total flow rate (regime II). In contrast, at low flow rates, the extent of increase in transport and thus r_t is insufficient to make up for the reduction in the effective surface area, resulting in $R_t < R_s$ (regime I).

Furthermore, the simulation data was used to inform scaling relationships between reaction rates and velocity and mass transfer coefficients. For cases when the liquid slugs are controlled by pore geometry, flow rate and fluid properties, $r_t \propto U^{0.45}$, whereas in single-phase flow $r_s \propto U^{0.1}$. In these cases, A_e/A_c is roughly the average saturation. However, when the liquid slugs are decoupled from the pore geometry and velocities, as would be expected for pores in a porous medium, the scaling laws change, and A_e/A_c can be larger depending on velocities etc. These observations have important implications for upscaling the impacts of two-phase flow dynamics on mineral reaction rates, and highlight the need for further research on transport behaviors and effective surface area for heterogeneous reactions in porous media.

While heterogeneity in mineral surface reactivity and roughness at finer scale did not appear to be important in our study, they may influence the mineral-fluid interactions depending on the reactive transport conditions (Deng et al., 2018; Fischer et al., 2012; Lüttge et al., 2013). In particular, given the stochastic nature of mineral nucleation and expectedly higher variability in precipitation rates (Deng & Poonoosamy, 2026), future studies examine the impacts of multiphase flow on mineral precipitation rate are needed.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The softwares used for the simulations are open-sourced and can be downloaded through a public Zenodo repository at X. Wang and Deng (2026).

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